Removing the Model from the Query Path: A Deterministic Graph-QA System as a Measurement Instrument, and What It Reveals About LLM Graders

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Abstract

Almost every question-answering system over knowledge graphs places a language model between the question and the database: the model writes a query, the query runs, and an answer is composed from the result. That design makes a class of controlled comparisons ill-posed, because a stochastic query-writing step sits between every manipulation and every score.

We describe a system that removes it. Ariadne answers natural-language security questions over two live knowledge graphs that are never merged and that speak different query languages – MITRE ATT&CK in Neo4j (Cypher) and MITRE CAPEC in Apache TinkerPop (Gremlin) – and on every shipped path *no language model writes a database query.* We use the resulting determinism as an instrument.

Our primary methodological finding concerns the instrument used to score such systems at all. Grading models have a *resolution floor*, and near the ceiling most cannot represent the effect sizes this literature reports. Of three local graders scored on answers of known quality, one emitted a single distinct value in the near-ceiling band, one emitted two, and one of those two *inverted* the ordering of mediocre and good answers in exactly the score range where our systems sit. Correlation against human labels – the statistic the field reports – cannot detect this.

We then run four controlled comparisons the architecture makes possible, three of which are null or negative. Hand-written query templates do not hold the system's accuracy: a schema-derived, template-free path matches them on the schema they were written for and exceeds them by +0.1684 judged correctness on 288 held-out questions. Templates nevertheless beat a model writing the queries (+0.0822), a result that is positive under both grader families and statistically solid under only one – and we can say *which* one only because the instrument was characterised first. Federating across two engines buys no accuracy over merging (−0.0334 and +0.0000), though building the merged store required rewriting 1,083 of 9,138 nodes while federation rewrote none. Finally, benchmarks co-authored with the system under test overstate it.

We report what the design costs: exact-record identity falls from 1.0000 to 0.0000 across five paraphrase tiers. We release the full experimental record, including six experiments retained because an earlier measurement of ours was wrong.

1 Introduction

The architecture in this paper is not the contribution, and saying so early is what makes the rest defensible. Mediator–wrapper federation is twenty-five years old. Templates rather than free generation is a named family in knowledge-base question answering. Attributing a failure to its earliest failing stage is published

prior art, as is read-only enforcement on generated graph queries. <D-027> A reviewer who suspects we claim any of these is right to reject.

What is unoccupied is the *measurement*. Consider the question: do hand-written query templates carry a graph-QA system's accuracy? Answering it requires two systems identical everywhere except in how the query is produced. In an architecture where a model writes the query you cannot build the second: the stochastic step remains between the manipulation and the score. The comparison is not difficult, it is ill-posed. A system that generates no query with a model is therefore a different *instrument*, not merely a different implementation.

Using that instrument taught us something we did not set out to find, and which we now believe is the more portable result. Every judged number in this literature, ours included, is produced by a language model scoring another model's output. The field validates those graders by correlation against human labels. Correlation is the wrong statistic. A grader can correlate well overall and still be unable to represent a 0.03 difference in the band where a paper's headline sits; worse, it can *reverse* the ordering inside a band while leaving the global correlation intact. We measured both failures in local graders we had been using, and we found them only because a near-miss forced the check.

Contributions.

1. A calibration procedure for LLM graders that reports *resolution* and *inversion* rather than correlation, and the finding that two of three local graders cannot adjudicate the effect sizes we intended to publish (§5). [EXP-2026-09-008]
2. A federated graph-QA system dispatching native queries in two graph query languages to two live, independently governed stores, with no model-written query on any shipped path (§3). <C-007>
3. Four controlled comparisons that architecture makes possible, three of them null or negative (§6–§9).
4. An accounting of what the design costs off its own vocabulary (§11), and of two scoring faults we deliberately did not fix (§13).

We claim priority on implementing and measuring. Nothing more.

Relation to TMLR's criteria. This submission contains no state-of-the-art result and does not claim architectural novelty. Three of its four system-level findings are null or negative. We submit it because its claims are, we believe, supported by the evidence presented, and because the grader result in §5 bears on how anyone in this audience validates an LLM-judged number.

2 Related work

[DRAFT NOTE – BLOCKED ON PROSE, NOT ON READING. The literature sweep closed 2026-09-06: CEUR-WS enumerated at 12,764 rows, four primary template systems read in full rather than by abstract, and an arXiv field sweep run with a validated control. Ten nearest neighbours are tabulated in the repository's `RELATED_WORK.md`. This section must discharge six specific obligations, each recorded with its reason:

1. Cite Emonet et al. (2024) ourselves as the single-language federated predecessor, before a reviewer finds it. Our federation claim rests *entirely* on the two-query-language axis and does not survive being stated more broadly. <C-024>
2. Carry the TITAN acknowledgement inline. We must never write that no ATT&CK question resource exists. <C-025>
3. Cite QAnswer's unswept abstention threshold – the nearest case of the field holding the relevant knob and never turning it.

4. Correct two citation errors inherited from secondary sources: QUINT is WWW 2017, and TeBaQA's substantive paper is SEMANTiCS 2021.

5. A defensive sentence separating *federated learning* from federated query execution. The homonym floods this literature.

6. Engage Unverified (2026) and PhantomBench directly: both validate the fabrication mechanism of §11 without occupying the graph-QA setting.

3 System

Five stages: **plan** → **route** → **retrieve** → **verify** → **compose**. A planner decomposes a question into sub-questions and assigns each to a source. Per-source specialists emit a native query in that source's own language – Cypher against Neo4j for ATT&CK, Gremlin against TinkerPop for CAPEC. Results are joined on shared identifiers over a declared bridge. A verifier admits or rejects each candidate claim against retrieved records. A composer writes the answer.

Where the model seams are. A language model may plan; it may grade, in evaluation only; and it may compose prose. **A language model never writes a query that reaches a store.** <C-007> The `llm_query` arm of §7 exists solely as a comparator and is not a shipped path.

Which arms narrate, stated precisely. The composition seam is where the arms differ, and the distinction is load-bearing for §13. Each answer artifact records its own configuration:

Arm	Query path	Composition (`run_meta.synthesis`)
`deterministic`	model-free	`deterministic` — model-free
`generic`	model-free	`llm (claim-grounded)` — **a model composes**
`llm_query`	a model writes the query	`llm (claim-grounded)`

So "no model writes a query" is exact, and "the system is deterministic" is not: `generic` — the arm carrying the primary result of §6 — composes its prose with a language model, and its judged figures therefore carry a run-to-run variance we report as unmeasured in §13. Only `deterministic` is model-free end to end, and that is verified rather than asserted: two full runs over the 284-question population are byte-identical in every field the scorer reads, differing only in wall-clock duration. [EXP-2026-09-010]

The verifier, ablated. Each of the verifier's four claim gates was disabled individually over 888 candidate claims. Disabling any single gate admitted *zero* additional unsupported claims. [EXP-2026-08-014] We report this as a weaker result than it sounds: on this population the gates are mutually redundant, not individually load-bearing. <C-017>

Stores. The CAPEC side loads 6,374 vertices and 8,582 edges across 25 node types and 34 edge types. [EXP-2026-09-001]

4 Experimental design

Arms. Three, differing only in query production.

Arm	Query produced by
`deterministic`	Hand-written parameterised templates
`generic`	Assembly from the graph schema at runtime; no templates
`llm_query`	A language model, given schema, resolved seeds, gate constraints, repair loop

Question populations. Nine, of which four carry weight here: a development catalogue; an expansion bank built from a schema census so coverage is structural rather than convenient (572 questions) [EXP-2026-09-001]; a 288-question holdout split off and sealed by SHA-256 before any repair was written [EXP-2026-09-007]; and 182 questions taken verbatim from two published external question sets, neither authored with sight of this code. [EXP-2026-08-010]

Grading. Every judged figure was produced under *two named grader families*, each identified by exact model id, quantisation and date – never by relay alias. This is not fastidiousness. One alias in this project served three different models over its lifetime, and a figure attributed to an alias is not attributable at all. <D-028>

Statistics. Paired percentile bootstrap, 2,000 resamples, seed 1729, 95% intervals throughout.

Two scoring faults we did not fix. Abstention boilerplate credits a refusal, and the scorer recognises the source's identifier vocabulary rather than the graph's (0.185 on the subset where it was quantified). **Both inflate scores.** Fixing either partway through a comparison would have invalidated the comparison, so both are priced into every figure below and restated in §13.

5 Grading models have a resolution floor

Why correlation is the wrong statistic. A grader is an instrument. An instrument is characterised by what it can *resolve*, not by how well it agrees on average. We therefore ask a narrower question than agreement: holding prompt and protocol fixed, on answers whose true quality band is known by construction, *how many distinct values does this grader emit in that band, and can it therefore represent the effect we intend to publish?*

The acceptance criterion was pre-registered before any judgment existed: a grader is usable near the ceiling iff fraction at $1.0 < 0.95$ and distinct values > 2. [EXP-2026-09-008]

Table 1: Near-ceiling resolution, $n = 20$ answers per grader. [EXP-2026-09-008]

Grader	Mean	Distinct	Values emitted	Usable
`gemma4-31B-mtp`	1.0000	**1**	$\{1.0\}$	no
`gemma-4-26B-it`	0.9750	**2**	$\{0.5, 1.0\}$	no
`qwen38-27B-v2`	0.9800	3	$\{0.8, 0.9, 1.0\}$	**yes**

A grader emitting one value in a band reports a contrast of exactly zero in that band *whatever is true*. This is not sampling noise; more samples do not help. It is a floor, and it rises as answers approach the ceiling.

Inversion, which is worse. `gemma-4-26B-it` does not merely lose resolution. Scored across constructed bands it places the *low* band at 0.6435 above the *mid* band at 0.6239, when the constructed ordering is mid above low. [EXP-2026-09-008] That is precisely the 0.40–0.65 range in which all three of our arms sit on the holdout. A grader that adds noise is survivable. A grader that reverses an ordering is a different failure, and it is *invisible to the correlation statistics the field reports*. We halted a planned third-family grading pass on this evidence, with zero judgments written.

What this cost us, reported as a limitation. Our co-authorship-bias effect is 0.0322 under the one grader that can resolve it and 0.0000 under a grader that cannot. [EXP-2026-09-006] <C-021> The second number reads as a refutation and is an instrument limit. Six of twelve cells in our cross-family re-grade are saturated under the second family. Separately, a tie-handling defect in our roll-up – excluding pairs tied by the reference grader but not pairs tied by the candidate – was one revision away from publishing *four arm-order reversals that do not exist*. The correct count is zero. [EXP-2026-09-006]

What it bought us. Because the instrument was characterised, we can state which of the two grader families in §7 to believe, rather than averaging them. That is the entire argument for doing this.

6 Templates versus schema-derived construction

This result replaces a claim our own experiments refuted: we previously held that the hand-written templates were where the accuracy lived. <C-003>

On the catalogue the templates were written for, the two paths are indistinguishable. Across ten anchor-disjoint banks, templates score 0.99 ± 0.0225 and the schema-derived path 1.00 ± 0.0000; paired difference -0.01 ± 0.0225. [EXP-2026-08-015] Templates do not hold the accuracy; on home ground they merely match it.

On a schema nobody wrote templates for, the ordering does not narrow – it inverts. Over the same ten banks on an identity schema: templates 0.2392 ± 0.0674, schema-derived 0.8514 ± 0.0890, paired difference -0.6121 ± 0.0981. [EXP-2026-08-015] On a blind set over that schema, retrieval is 0.1964 for templates against 0.7857 derived. [EXP-2026-08-011]

The failure mode matters more than the gap. On the foreign schema the template arm did not decline. It asserted: unsupported-claim rate 0.8214 against the derived path's 0.1071. [EXP-2026-08-011] A template that does not fit the schema still fires, still retrieves, and still answers confidently. <C-023>

Table 2: Judged correctness, schema-derived minus templates, on unseen questions.

Population	n	Gap	Source
Expansion bank	284	+0.1642	[EXP-2026-09-002]
Expansion bank, post-repair	284	+0.1623	[EXP-2026-09-005]
Sealed holdout	288	+0.1684	[EXP-2026-09-007]
Sealed holdout, second grader family	288	+0.2143	[EXP-2026-09-007]

The sharpest single piece of evidence in the paper. We diagnosed and repaired two retrieval defects in the template arm. Template retrieval rose by +0.0684. Template judged correctness moved +0.0012. [EXP-2026-09-005] Retrieving the right records did not make the answers right. Whatever the templates cost, it was not recall.

A weighting caveat we insist on. The census-weighted repair gain is +0.0017, against +0.0503 pooled. [EXP-2026-09-007] The pooled figure is an artifact of a bank that deliberately over-samples rare question shapes. Both must be quoted, in that order.

7 Templates versus a model writing the queries

This is the weakest of our three system comparisons and we do not lead with it.

Given the real schema, resolved seed entities, the gate's constraints and a repair loop, a model writing the queries scores 0.4039 judged correctness on the holdout, against templates at 0.4862 and the schema-derived path at 0.6546. [EXP-2026-09-007] On a fresh bank it scored 0.5000 against templates at 0.9467, [EXP-2026-08-008] and it emitted an empty query on 153 of 284 expansion-bank questions. [EXP-2026-09-002] <C-012>

Table 3: Templates minus model-written queries, judged correctness, 95% paired bootstrap intervals, under two grader families. [EXP-2026-09-006] [EXP-2026-09-007]

Half	Grader family 1 (`qwen38-27B-v2`)	Grader family 2 (`gemma4-31B-mtp`)
Development	+0.1147 [+0.0653, +0.1645]	+0.0696 [−0.0024, +0.1450]
Holdout	+0.0822 [+0.0362, +0.1308]	+0.0643 [−0.0080, +0.1333]

Four measurements, four positive point estimates, two intervals excluding zero and two not. We report this as *positive under both grader families and statistically solid under one.* §5 is what lets us name which family that is, instead of averaging two instruments of unequal resolution and reporting the mean as though it meant something.

8 Federated versus merged execution

We built both conditions, changed one variable, and used no model in either arm. Merged minus federated judged correctness: −0.0334 on one blind bank, +0.0000 on a second; mean −0.0167. [EXP-2026-08-009] Federating bought no accuracy. We state that as a result rather than working around it. <C-015>

What survives the null is structural and countable. Constructing the merged store required rewriting 1,083 of 9,138 nodes – 22.71% of the CAPEC side, 11.85% of the whole store – including re-prefixing 189 taxonomy entries. Federation rewrote zero. [EXP-2026-08-017] Merging is not free; it is a data-editing cost paid once per source and again on every refresh. <C-022>

A claim we deliberately do not make. A third graph was loaded unmodified beside both catalogues. It is tempting to say it could not legally have been copied into a merged store. Our own record forbids that claim: the contractual argument is a motivation, the identity records are synthetic, and what was demonstrated is narrower – keeping the sources separate asked that graph for nothing.

An apparent contradiction we owe the reader. SPLIT-RAG reports a 17.6 Hit-point gap between split and merged conditions where ours is neutral. The most likely reconciliation is that their ablation varies agent count alongside store topology, making it a different variable. We state this ourselves rather than leaving it for a reviewer.

9 Co-authored benchmarks overstate their systems

Evaluating a system on questions written alongside the code that answers them inflates the result. Measured on the same system, same grader, same session: the gap is 0.1472 before a composer fix [EXP-2026-08-007] and 0.0483 after, [EXP-2026-08-008] and 0.0322 under the named, resolution-checked grader. [EXP-2026-09-006] <C-021>

The bias's dominant form is not a wrong answer. It is a question the bank does not contain. This is why §2's external question sets matter more than their size suggests.

10 Failure localisation

[DRAFT NOTE – BLOCKED. Open as Q24 since 2026-08-26: our failure distribution runs opposite to the only comparable published distribution, and we do not yet have a written account of why. This is a day of analysis, not an experiment, and it is the longest-standing item on our critical path.]

The material this section already has: on 182 externally-authored questions, an oracle plan reaches 0.9863 micro-recall while the template arm reaches 0.6836 and a plan-free model reaches 0.6132. [EXP-2026-08-010] Eight of fourteen question families match the ceiling; *four return nothing at all.* The schema-derived path on the 60-question subset reaches 0.9953, essentially the oracle. The failure is concentrated and localisable rather than diffuse – which is what makes the contradiction worth explaining rather than waving at. <C-016>

11 The coverage cliff

The control confirms the harness: on unmodified questions the system is exact. Two tiers of paraphrase halve it. By T4 it is zero, and excluding catalogue dumps the recall at T4 is 0.0013. At T3 the system responds to 52.48% of questions with a catalogue dump – retrieving broadly rather than answering. <C-014>

Table 4: Exact-record identity across five paraphrase tiers of the same questions. [EXP-2026-08-020]

	T0 (control)	T1	T2	T3	T4	T5
Records exact	**1.0000**	0.4667	0.4250	0.0990	**0.0000**	**0.0000**

It answers about entities that do not exist. On 30 questions naming entities with no referent, the deterministic arm answered 4, with up to 558 genuinely retrieved and genuinely cited records behind them. [EXP-2026-08-020] One arm abstained on 93.33% of them, which is the existence proof that the behaviour is fixable rather than intrinsic.

Reporting this trade, rather than a victory, is what inoculates the paper against its own honest finding.

12 Run-to-run variance, where it lives and what it costs

Every judged figure in this paper is one run of one arm. We measured whether that matters, separately for each seam.

The query path reproduces exactly. Two independent runs of both model-free arms over the same 284 questions return byte-identical retrieval, coverage, claims and evidence. [EXP-2026-09-010] This is the first direct measurement of the determinism the architecture is supposed to buy, and it holds.

The composition seam does not. `generic` narrates with a language model, and 143 of 276 comparable answers are worded differently on a rerun at the same seed. [EXP-2026-09-010] `deterministic`, which composes without a model, is identical on all 284.

Seed choice barely matters for the query-writing arm. Across three seeded runs, the across-seed spread of `llm_query` is 0.0089, an order of magnitude below the 0.0822 margin of §7. [EXP-2026-09-009]

[DRAFT NOTE — the remaining cell is running: how far `generic`'s judged score moves across seeds. Until it reports, §6's figures are quoted as point estimates with the caveat in §13, and this section states the bound rather than the value.]

13 Limitations

Every item here is measured. Stating them is cheaper than being caught.

1. **The templates-over-generation result is grader-dependent** (§7): solid under one family on both halves, not solid under the other on either.
2. **No external system baseline.** Every comparison is between arms of this system. We have 182 externally-authored *questions* – an outside anchor on the inputs, not on the systems. This is the single largest gap and we do not minimise it.
3. **No human evaluation.** Every correctness number is a model grading a model. §5 characterises that instrument; it does not substitute for a human anchor.
4. **The holdout is a random split of a bank we authored.** It controls for tuning to particular questions. It does *not* control for the co-authorship bias we ourselves measure in §9.
5. **Several experiments were run once**, with seeds unrecorded, and this applies to the arm carrying the primary result. `deterministic` is exempt by measurement: it is byte-identical across two full runs in every scored field, so a single run of it is sufficient by construction. [EXP-2026-09-010] **`generic` is not exempt, and this is measured rather than suspected**: on a rerun at the same seed,

against the same store, **143 of 276 comparable answers are worded differently**. [EXP-2026-09-010] Its figures in §6 are therefore one wording of that arm, not a point estimate of it.

Two things bound how bad this is. First, the variation is confined to *one field*: every retrieval, coverage, claim and evidence field is byte-identical across runs, so what varies is the prose and not what the system found. Second, a different wording need not be a different score, and how much of this reaches the judged number is a separate measurement we report in §12. We state the unbounded version here because it is the honest one until that measurement is quoted.

6. **The query-writing arm is stable across seeds, so its result is not a single-draw artifact**: three seeded runs give an across-seed spread of 0.0089 against a margin of 0.0822. [EXP-2026-09-009] We record this because the natural objection to §7 — that one draw established nothing — is one we tested and can answer.

7. **Two scoring faults left unfixed by design** (§4). Both inflate scores.

8. **Weighting matters**: the census-weighted repair gain is +0.0017 against +0.0503 pooled.

9. **One domain, two schemas, two query languages.** Nothing here is a law about query construction until it replicates where we wrote no templates.

10. **Source coverage is partial**: 12 of 34 edge types reached, and 0 of 12 facet properties used at census time. [EXP-2026-09-001]

11. **A reserved-word Gremlin binding** costs the model-written-query arm evidence on some questions. Pre-existing and priced into the figures, but whether it originates with us or with the model is undiagnosed.

Broader Impact Statement

The system answers questions over public cybersecurity taxonomies. Two impacts are worth stating.

First, §11 shows a retrieval-grounded, citation-producing system answering confidently about entities that do not exist, with hundreds of genuinely retrieved and genuinely cited records behind the answer. Citations are routinely read as evidence of grounding. In a security setting, a fluent, cited answer about a technique that was never published is a plausible route to a bad operational decision. We report the failure rather than tuning it out of the evaluation, and we regard the ability to decline with a machine-checkable proof as the more important property of the design.

Second, §5 implies that some published differences between LLM-judged systems may be below the resolution of the judge that measured them. We make no claim about any specific published result. We note only that resolution is not currently reported, that correlation does not reveal it, and that we found two graders in our own pipeline that could not adjudicate the effect we intended to publish.

References

V. Emonet et al. [verify] single-language federated question answering over knowledge graphs. *arXiv preprint arXiv:2410.06062*, 2024. v4. MUST be cited by us as the single-language federated predecessor. Our federation claim rests entirely on the two-query-language axis.

Unverified. [verify] deceptive grounding. *arXiv preprint arXiv:2607.09349*, 2026. Reports 8–87% attribution failure across 13 models in clinical RAG; its ablation states that removing entity-specific evidence shifts all failures to confabulation. Validates the mechanism of our Section on the coverage cliff without occupying the graph-QA setting.

A Record defects outstanding at submission

Found by an independent readiness audit on 2026-09-06 and not yet closed. Each is checkable by a reviewer; we list them rather than wait to be asked.

partly closed There was no check that numbers in this paper trace to an experiment, despite an internal claim that one existed. One was written 2026-09-07 and reports 17 untraced numbers, of which twelve were the confidence intervals in Table 2 — now recorded as manifest metrics. It is a review list, not a gate, and it does not run in CI.

closed The grader-calibration experiment underpinning §5 was marked *planned* with an unwritten results file while this draft rested a section on it. Written up and marked complete 2026-09-06; all four of its pre-registered hypotheses hold. [EXP-2026-09-008]

closed Four of the holdout experiment's recorded metrics were taken from an intermediate roll-up and were wrong — including one third-decimal error of 0.009, larger than an earlier draft of this appendix described. All four were adjudicated by recomputing from the 3,456 raw judgment sidecars and corrected 2026-09-07. No conclusion moves: every value keeps its sign and its significance verdict. [EXP-2026-09-007] The intermediate file declared its own incompleteness in a field named `not_measured`, and every wrong value lay inside what that field named — so the error was disclosed when it was made and survived anyway, which locates the defect in the reading rather than the writing.

1. One experiment is marked superseded, not six. Any statement about retaining our mistakes must say one.
2. Withdrawn figures remain in repository prose and must be retracted in place.
3. Seeds are unrecorded on several runs; three seeded repeats of the model-written-query arm are owed.
4. An internal protocol asserted that both the `deterministic` and `generic` arms were model-free and could therefore be run once. The artifacts contradict that for `generic`, whose composition seam calls a language model. The assertion was never checked against a single artifact before it was relied on, and the run-to-run spread of the arm carrying the primary result is consequently still unmeasured. [EXP-2026-09-010]

B Traceability

Per our internal standing rule, no number appears in this paper without an experiment identifier. The `[EXP-...]` tags throughout are those identifiers; `<C-...>` and `<D-...>` tags reference the project's claim and decision ledgers, which are released with the artifact. Set `\togglefalse{trace}` in the preamble to typeset a clean reading copy.